R&S SwissQual AG®··· NetQual
Call Analysis Description
Manual
Release 20.3

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Version 20.3

ROHDE&SCHWARZ
Make ideas real

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1 Introduction

You can run the Call Analysis script in a R&S SwissQual NQDI 20.3 post-processing application.

1.1 About NQDI

NQDI is a post-processing system maximizing the potential of data that is collected by QualiPoc and Diversity products, for network and service optimization, and benchmarking. Based on data from all radio technologies, NQDI provides automated data validation, detailed quality analysis and troubleshooting, and long-term reporting for voice and data services.

NQDI is designed as a client/server application that uses Microsoft SQL Server as a repository for the measurement data. The client application system includes features for data administration, filtering capability, analysis, statistics, map plotting, and KPI report generation.

1.2 Overview

The Call Analysis script runs several SQL scripts to analyze call parameters, such as, call sequence and radio coverage, in the measurement files that are in your database. At the end of the process, the Call Analysis script generates a report of the analysis in Microsoft Excel format. The Call Analysis script analyses UMTS and GSM calls in detail. However, for CDMA calls the script only analyses radio coverage and the speech quality of other technologies.

The Call Analysis is executed for both devices involved in a **mobile to mobile** scenario to figure out if a failed or a dropped call is caused by the A-side or the B-side device

The "CallAnalysis" feature consists of the following parts:

- "CallAnalysis": Database table that contains the results
- "spCallAnalysis": Stored procedure that scans a call session and collects the relevant information
  The "spCallAnalysis" procedure must run once in each call.
- "spCallAnalysisCode": Stored procedure that calculates the "Call Analysis Code" for each call
  This procedure uses the results that the "spCallAnalysis" stored procedure returns.
- "Session Analysis Settings": Dialog box in NQDI

► To access the "Session Analysis Settings" dialog box, click "Session Analysis" on the "Configuration" menu.

- `CallAnalysis.XLS` report file in Excel format

► To generate the `CallAnalysis.XLS` report, click "Reports" on the "Tools" menu, point to "Voice", and then click "CallAnalysis".

2 Call Analysis Script

You can run the Call Analysis script manually in NQDI or you can specify that NQDI automatically runs the script after you import a measurement file.

► To manually run the "Call Analysis" script in NQDI, click "Session Analysis" on the "Configuration" menu, and then click "Recalculate" on the "Options" tab.

► To specify that NQDI runs the script automatically after you import a measurement file, click "Session Analysis" on the "Configuration" menu, and then click "Run Call Analysis script automatically after import".

The Call Analysis script takes 10 to 15 seconds to process each call.

2.1 Session Analysis Settings

► To specify the settings for the Call Analysis script in NQDI, click "Session Analysis" on the "Configuration" menu.

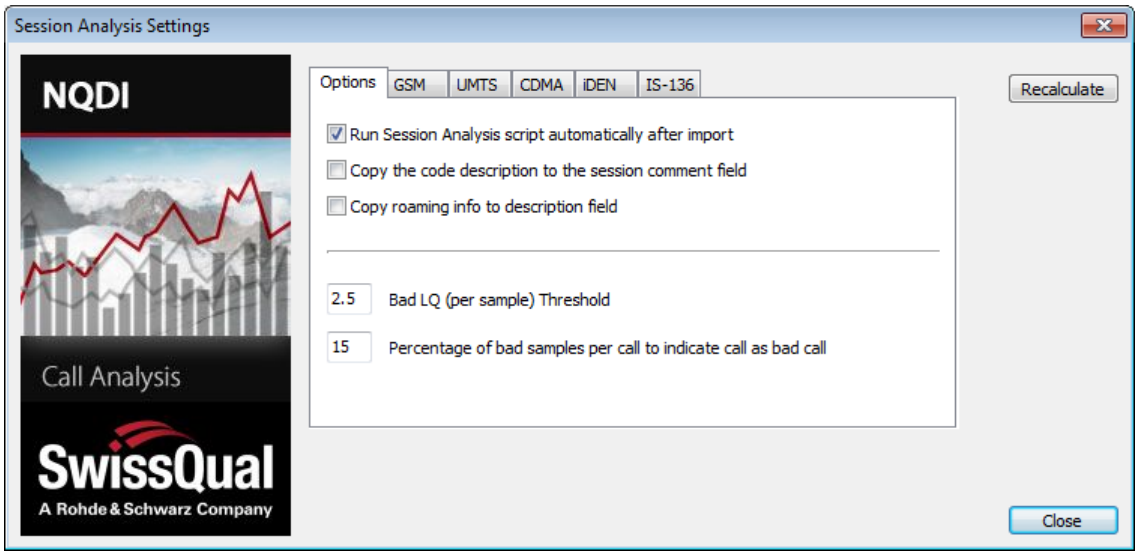

Figure 2-1: Options tab in Settings dialog box for the Call Analysis script

If you want to manually run the script with your new settings, click "Recalculate" on any tab in the "Session Analysis Settings" dialog box.

2.1.1 Options Tab

Table 2-1: Description of settings on the Options tab for the Call Analysis feature

Setting	Description
"Run Session Analysis script automatically after import"	Runs the "Session Analysis Process" script automatically after NQDI imports a measurement file. After the script runs the "sqCallAnalysis" stored procedure on each new call in the database, the "sqCallAnalysisCode" procedure calculates the Code and the Code Description. This process can take up to 15 seconds to complete for each call.
"Copy the code description to the session comment field"	Automatically copies the code description text to the comment field of a session.
"Copy roaming info to the description field"	Automatically copies the roaming info for a call to the description field of the call.

2.2 Threshold Value Options

Diversity measures two types of threshold values, namely coverage values speech quality values.

2.2.1 Speech Quality

Speech threshold values use the Mean Opinion Score (MOS) scale, which ranges from 1 to 5. For example, if a speech test yields an LQ value that is below the set threshold value, the call contains a bad sample. If the percentage of bad samples exceeds the threshold value that you set in NQDI, the call is a Bad Call. The default value for a bad sample is 2.5 on the MOS scale and the percentage of bad samples that result in a bad call is 15%.

2.2.2 Signal Strength and Quality

Diversity evaluates signal strength and signal quality for GSM, UMTS, and CDMA technologies, each of which has different signal indicators.

This evaluation is only performed on failed calls and dropped calls.

If the measurements exceed or fall below the threshold values, the description of the code is qualified as "High" or "Low" respectively, for example, "High RxQual" or "Low RxLev".

2.2.3 GSM Tab

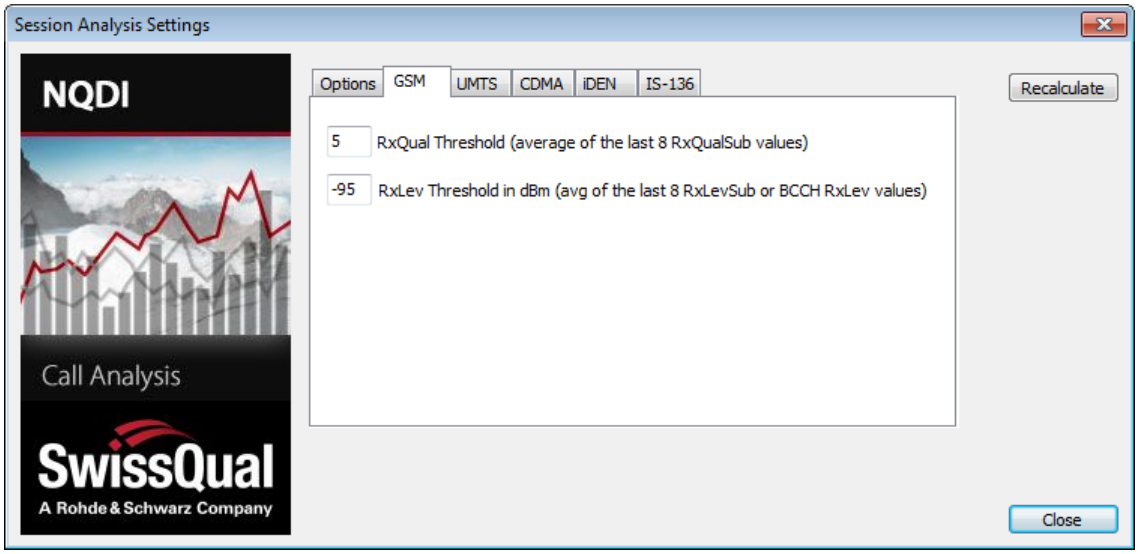

Figure 2-2: GSM tab in Settings dialog box for the Call Analysis script

Table 2-2: Description of settings on the GSM tab for the Call Analysis feature

Setting	Description
"RxQual Threshold (average of the last 8 RxQualSub values)"	If the call analysis script returns a value that is higher than the threshold value in this box, the call has a "High" RxQual level. The default value is 5.0
"RxLev Threshold in dBm (average of the last 8 RxLevSub or BCCH RxLev values)"	If the call analysis script returns a value that is less than the threshold value in this box, the call has a "Low" RxLev value. The default value is -95 dBm.

2.2.4 UMTS Tab

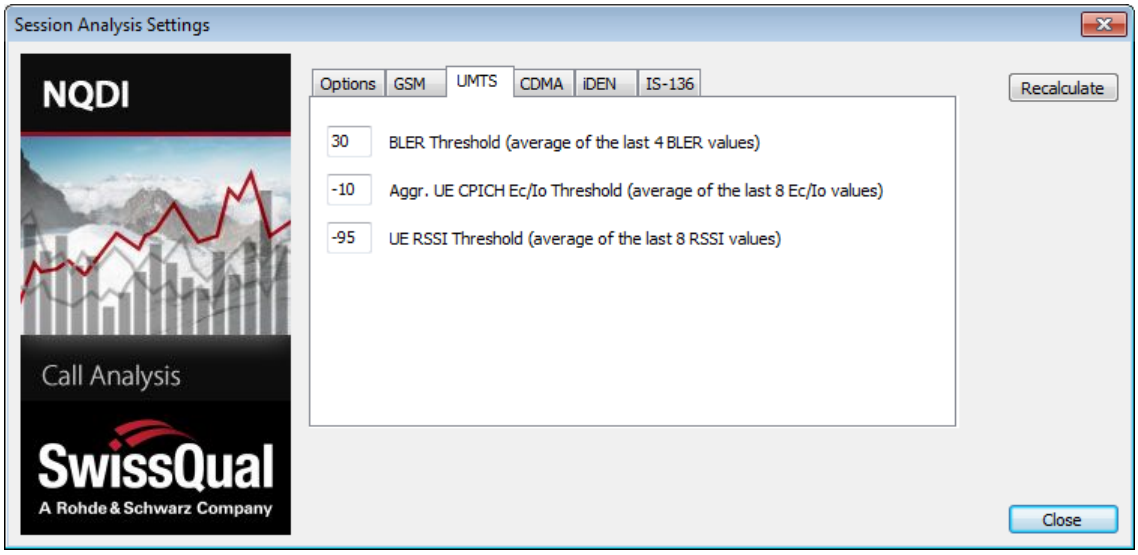

Figure 2-3: UMTS tab in Settings dialog box for the Call Analysis script

Table 2-3: Description of settings on the GSM tab for the Call Analysis feature

Setting	Description
"BLER Threshold (average of the last 4 BLER values)"	If the call analysis script returns a value that is less than the threshold value in this box, the call has a "High" BLER value. The default value is 30.0
"Total Ec/Io Threshold in dB (average of the last 8 aggregated Ec/Io values)"	If the call analysis script returns a value that is less than the threshold value in this box, the call has a "Low" Total Ec/Io value. The default value is -10 dB.
"UE Rx Power Threshold in dBm (average of the last 8 UE Rx Power values)"	If the call analysis script returns a value that is less than the threshold value in this box, the call has a "Low" UE Rx Power value. The default value is -95 dBm.

2.2.5 CDMA Tab

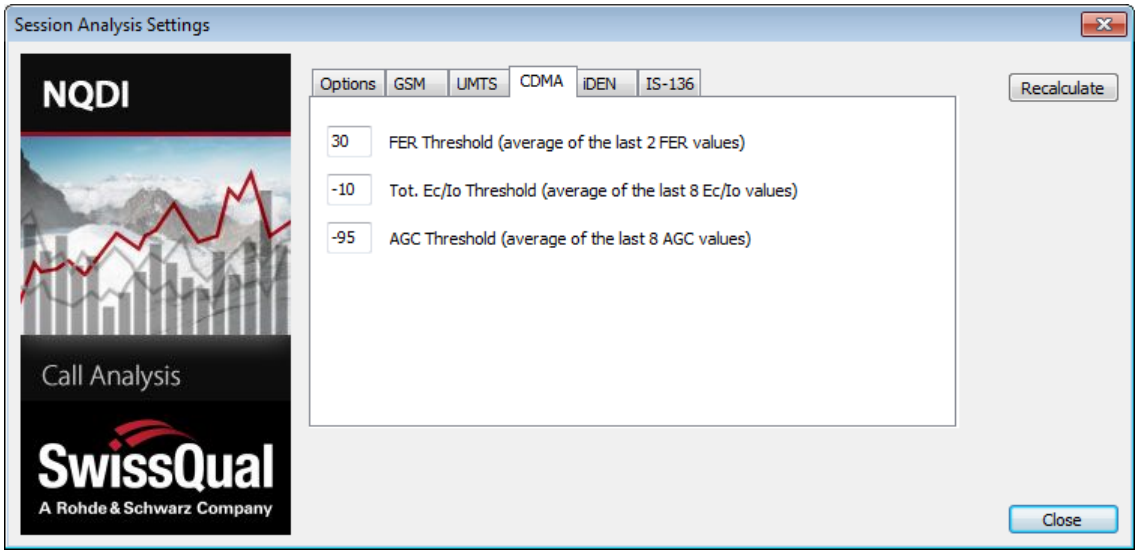

Figure 2-4: The dialog for CDMA related thresholds

Table 2-4: Description of settings on the CDMA tab for the Call Analysis feature

Setting	Description
"FER Threshold (average of the last 4 BLER values)"	If the call analysis script returns a value that is less than the threshold value in this box, the call has a "High" FER value. The default value is 30.0
"Total Ec/Io Threshold in dB (average of the last 8 aggregated Ec/Io values)"	If the call analysis script returns a value that is less than the threshold value in this box, the call has a "Low" Total Ec/Io value. The default value is -10 dB.
"AGC Threshold in dBm (average of the last 8 UE Rx Power values)"	If the call analysis script returns a value that is less than the threshold value in the box, the call has a "Low" AGC value. The default value is -95 dBm

2.2.6 iDEN Tab

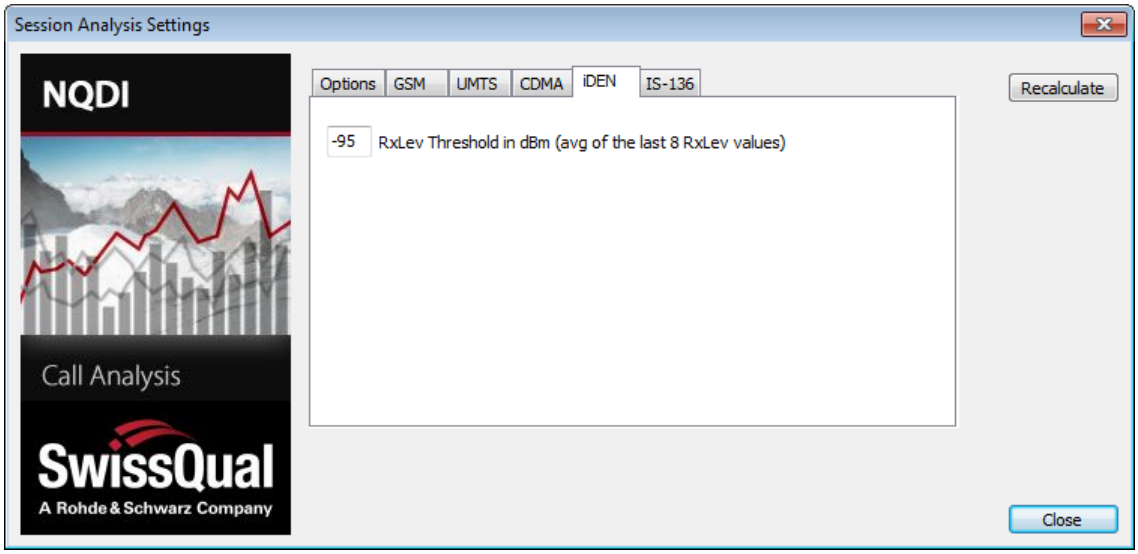

Figure 2-5: The dialog for the iDEN receive level threshold

Table 2-5: Description of settings on the CDMA tab for the Call Analysis feature

Setting	Description
"RxLev Threshold in dBm (average of the last 8 UE Rx Power values)"	If the call analysis script returns a value that is less than the threshold value in this box, the call has a "Low" iDEN RxLev value. The default value is -95 dBm

2.2.7 IS-136 Tab

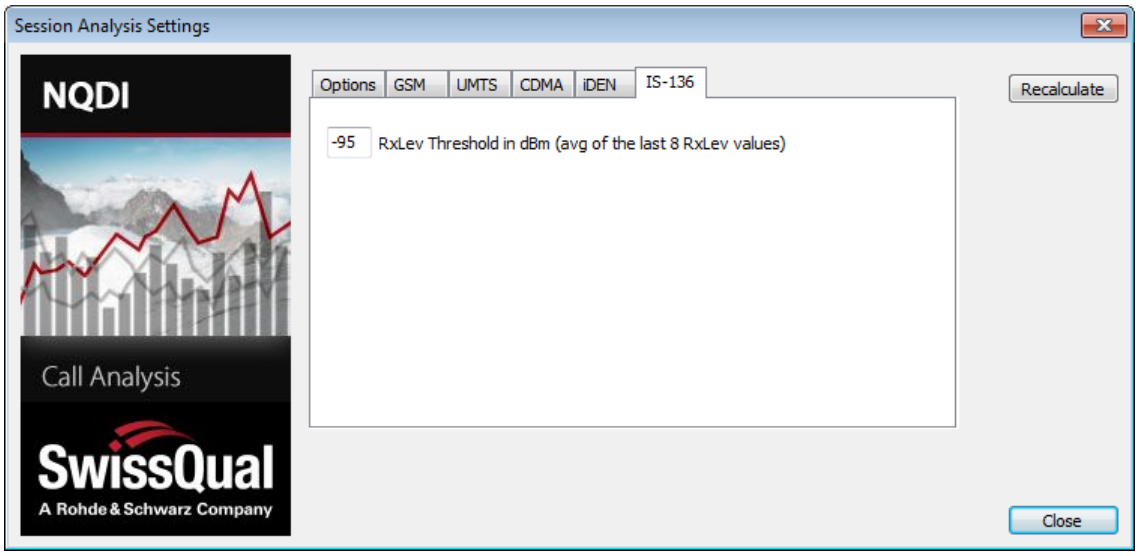

Figure 2-6: The dialog for the IS-136 RxLev threshold

Table 2-6: Description of settings on the CDMA tab for the Call Analysis feature

Setting	Description
"RxLev Threshold in dBm (average of the last 8 Rx Power values)"	If the call analysis script returns a value that is less than the threshold value in this box, the call has a "Low" IS-136 RxLev value. The default value is -95 dBm

2.3 Recalculating Values

When you change the threshold values, you do not need to completely rescan each call. Instead, you can recalculate the values from within the "Session Analysis Settings" dialog box.

► To recalculate the values, click "Recalculate" on any of the tabs in the "Session Analysis Settings" dialog box, and then click "Yes".

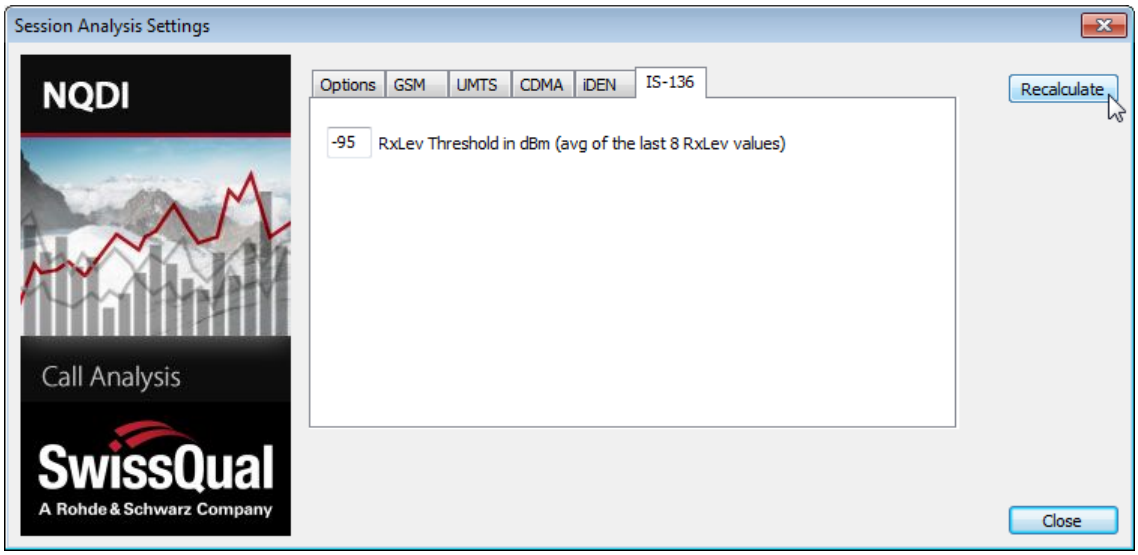

Figure 2-7: Recalculate button in CallAnalysis Settings dialog box

Note:

- You cannot run the call analysis script during an import procedure
- You cannot exit NQDI while you run the Call Analysis script
- You cannot import data while you run the Call Analysis script
- You cannot generate a Call Analysis report while you run the Call Analysis script

2.4 Analysis Comment on Analysis Tab

The call analysis script runs for both the A and the B-side device and the "Comments" tab under "Analysis" tab reports comments on the result:

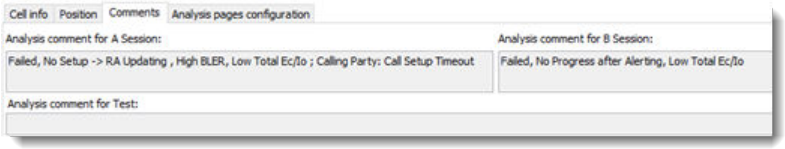

Figure 2-8: Comments on Analysis tab

3 CallAnalysis Table

Table 3-1 contains one row per call session. The "sqCallAnalysis" stored procedure calculates most of the parameters, while the "sqCallAnalysisCode" calculates the "Code" and "CodeDescription" fields.

Table 3-1: Description of CallAnalysis Table

Field Name	Type	Description
sessionId	int	Link to Sessions.sessionId
FileId	int	Link to FileList.FileId
NetworkId	int	Link to NetworkInfo.NetworkId
PosId	int	Link to Positions.PosId
callType	Varchar(5)	M->L or L->M
callDir	Varchar(5)	A->B or B-> A
callStatus	Varchar(50)	"Completed", "Failed", "Dropped" or "System Release"
setupTime	real	Time between "Dial" and "Connect", which is taken from CallSession.SetupTime
callCause	Varchar(300)	Last received Cause value from CC:DISCONNECT message where the value is taken from CallSession.CallCause
callTerminationType	Varchar(100)	Call termination code based on Seven.Five Statue machine. Taken from Call-session.CallTermination
valid	smallint	Flag indicating if this session is valid or not
retryTerminationType	Varchar(100)	Unused
technology	varchar (50)	Technolog(ies) that are used in the call
band	varchar (50)	Frequency band
Roaming	varchar (50)	"Yes" for roaming "No" for not roaming
NetIds	int	Number of different NetIDs in that session
callDuration	int	The time between "Dial" and "Call" end, which is taken from the "Markers" table where the difference between the "Dial" event and the "CallTerminated" event is calculated
callStartTimeStamp	Datetime2	Timestamp of the "Dial" Event
callEndTimeStamp	Datetime2	Timestamp of the "CallTerminated" Event
EndTechnology	varchar (50)	Technology at the end of the call
Layer3Messages	int	0 for no layer 3 messages in entire session 1 for at least on message
SgnChnReq	Datetime2	Timestamp of the first Channel Request
SgnChnAss	Datetime2	Timestamp of the first IMM_ASS message
alertTimeStamp	Datetime2	Timestamp of the first Alert message
connAckTimeStamp	Datetime2	Timestamp of the first CONN_ACK message

Field Name	Type	Description
callSetupEndTimeStamp	dateTime2	Timestamp that indicates the end of a call set up. The timestamps is either from the CON_ACK message in a successful case or from the call termination an unsuccessful case
CallDisconnectTimeStamp	Datetime2	Timestamp when the call is disconnected
StartTechnology	varchar (50)	Technology at the beginning of the call
avgRxQual	real	Average RxQualSub of the last 8 Layer1 messages before the end of the call
avgRxLev	real	Average RxLevSub of the last 8 Layer1 messages before the end of the call
avgTA	real	Average TA of the last 8 Layer1 messages before the end of the call
avgMsTxPwr	real	Average TxPwr of the last 8 Layer1 messages before the end of the call
AvgBCCHRxLev	real	Average BCCH_RxLev of the last 8 Layer1 messages before the end the call
numOfRadioValues	int	Number of averaged values. Equal or less than 8.
LastRadioTimeStamp	Datetime2	Timestamp of the last Layer1 message in this session
avgScanRxLev	real	Unused
numOfScanValues	real	Unused
lastScanTimeStamp	Datetime2	Unused
numBetterNeighbors	int	Number of Neighbours with stronger BCCH values
numBetterScannedGSM	int	Number of better scanned channels
NumUlSamples	int	Number of Uplink voice samples in this session
NumDlSamples	int	Number of Downlink voice samples in this session
NumBadSamples	int	Number of Samples below bad call threshold
percentage	real	Average LQ
numBetterNeighbors,	int	
numChnReq	int	Number of Channel Requests in this session. NQDI only counts the messages where an IMM_ASS message with a corresponding random code is found
numCMServiceRequest	int	Number of CMServiceRequests in this session
numCMServiceReject	int	Number of CMServiceRejects in this session
numpageResponse	int	Number of Paging Responses in this session
numImmASS	int	Number of IMM_ASS in this session
numImmAssRej	int	Number of IMM_ASS_REJ in this session
numSetup	int	Number of SETUP in this session
numCallProceding	int	Number of Call Proceeding in this session
numAssCmd	int	Number of ASS_CMD in this session
numAssCmpl	int	Number of ASS_CMPL in this session
numAssFail	int	Number of ASS_FAIL in this session
numProgress	int	Number of PROGRESS in this session

Field Name	Type	Description
numAlerting	int	Number of ALERTING in this session
numConn	int	Number of CONNECT in this session
numConnAck	int	Number of CONN_ACK in this session
numChnRelCallSetup	int	Number of Channel Releases for the call setup in this session
numLURequest	int	Number of LU Request in this session
numLUAccept	int	Number of LU Accept in this session
numLUReject	int	Number of LU Reject in this session
numDisconnect	int	Number of DISCONNECT in this session
numRelease	int	Number of RELEASE in this session
numRelCmpl	int	Number of REL_CMPL in this session
numChnRel	int	Number of Channel Release in this session
numDCCH2GTrial	int	Number of KPI 15700 measurements
numDCCH2GSuccess	int	Number of successful KPI 15700 measurements
numTCHTrial	int	Number of KPI 15600 / 15800 measurements
numTCHSuccess	int	Number of successful KPI 15600 / 15800 measurements
LastHoTimeStamp	Datetime2	Timestamp of the last Handover before session end
LastHoType	Varchar	Type of the last Handover before end of the session (ASS_CMD, ASS_CMPL, ASS_FAIL, HO_CMD, HO_CMPL, HO_FAIL)
LastHoCause	Varchar	RR Cause value of the Last HO
avgRLT	real	Unused
numOfRLTValues	int	Unused
LastRLTTimeStamp	Datetime2	Unused
configCall	tinyint	Indicates if this was a configuration call, that is, to start or to stop a scenario
NumReestablishmentRe-quests	tinyint	Number of Call Reestablishment Requests
disconDirection		Direction of the DISCON message (U=Uplink, D=Downlink)
disconClass		Class value of the Disconnect message
disconCause	Varchar	Class value of the Disconnect message
disconLocation		Location value of the Disconnect message
code	Varchar(255)	See Chapter 4, "Code and CodeDescription Fields", on page 22.
noL3Duration		Indicates the number of seconds, in 10 second steps, that layer3 messages were not received from the mobile station
codeDescription	Varchar	See Chapter 4, "Code and CodeDescription Fields", on page 22.
NoService	Smallint	Indicates the number of occurrences of situations where the mobile reported to be in SCAN mode, means out of service

Field Name	Type	Description
Initializing	Smallint	
LastAssTimeStamp	Datetime2	Timestamp of the last ASSIGNMENT message (ASS_CMD, ASS_CMPL, ASS_FAIL)
LastAssMsg	Varchar	Type of the last ASSIGNMENT message (ASS_CMD, ASS_CMPL, ASS_FAIL)
LastAssCause	Varchar	Cause value of the last ASSIGNMENT message
CRRandomCode	Char(2)	Random code of the IMM_ASS message that corresponds to the Random code of the Channel Request message
RRCConnReqTimeStamp	Datetime2	Number of Connection Request by UMTS
RRCConnSetupTimeStamp	Datetime2	Number of Connection Setup by UMTS
numMMTrial	int	Number of completed Connection Setups by UMTS
numMMSuccess	int	Number of Radio Bearer Setup
avgBLER	real	Average BLER of the last 4 messages before the end the call
avgTotEcIo	real	Average aggregated Ec/Io of the last 8 messages before the end the call
avgUETxPwr	real	Average UE TxPower of the last 8 messages before the end the call
avgUERxPwr	real	Average UE RxPower of the last 8 messages before the end the call
numOfWCDMARadioValues	int	Number of WCDMAAGC values used
LastWCDMARadioTimeS-tamp	Datetime2	Timestamp of the last WCDMAAGC message used
numBetterMonitored	int	Unused
numBetterDetected	int	Unused
numBetterScannedUMTS	int	Unused
numRRCConnReq	int	Unused
numUMTSpageResponse	int	Unused
numRRCConnSetup	int	Number of RRC Connection Setup messages in the call
numRRCConnRej	int	Number of RRC Connection Reject messages in the call
numRBSetup	int	Number of Radio Bearer Setup messages in the call
numRBSetupCompl	int	Number of Radio Bearer Setup Complete messages in the call
numRBSetupFail	int	Number of Radio Bearer Setup Failure messages in the call
numRRCConnRelCallSetup	int	Unused
numRRCConnRel	int	Number of RRC Connection Release messages in the call
numDCCH3GTrial	int	Number of KPI 16000 measurements
numDCCH3GSuccess	int	Number of successful KPI 16000 measurements
numRABTrial	int	Unused
numRABSuccess	int	Unused
LastUMTSHoTimeStamp	Datetime2	Timestamp of last IRAT Handover

Field Name	Type	Description
LastUMTSHoType	varchar(100)	Type of last IRAT Handover
LastUMTSCause	varchar(100)	Unused
numGMMMessages	int	Number of GMM messages in the call
numSMSMessages	int	Number of SMS messages in the call
numMMMessages	int	Number of MM messages in the call
RLCexp	int	1 if RLC expired in call 0 if not
avgCDMAFER	real	Average FER of the last 2 messages before the end the call
avgCDMATotEcIo	real	Average aggregated Ec/Io of the last 8 messages before the end the call
avgCDMATxPwr	real	Average TxPower of the last 8 messages before the end the call
avgCDMAAGC	real	Average AGC of the last 8 messages before the end the call
numOfCDMARadioValues	int	Number of FingerInfo values used
LastCDMARadioTimeStamp	Datetime2	Timestamp of the last FingerInfo message
numCDMAOriginationMsg	int	Number of Origination messages in the call
numCDMAPageResponse	int	Number of Page Response messages in the call
numCDMAServiceConnect	int	Number of Service Connect messages in the call
numCDMAServiceCon-nectCmpl	int	Number of Service Connect Complete messages in the call
numCDMAChnAssMsg	int	Number of Channel Assignment Messages in the call
numCDMAInitState	int	Unused
numCDMAOriginationState	int	Unused
numCDMAPageResponseS-tate	int	Unused
numCDMARegistrationState	int	Unused
numCDMAAuthentication-State	int	Unused
numCDMAWaitForOrder-State	int	Unused
numCDMAWaitForAnswer-State	int	Unused
numCDMAConversationState	int	Unused
numCDMAReleaseOrder	int	Number of Release Order messages in the call
avgTDMABER	real	Average BER of the last 8 layer1 messages before the end the call
avgTDMARxLev	real	Average RxLev of the last 8 MsgIS136Report messages before the end the call
avgTDMATxLev	real	Average TxLev of the last 8 MsgIS136Report messages before the end the call
avgTDMATA	real	Average TA of the last 8 MsgIS136Report messages before the end of the call
numOfTDMARadioValues	int	Number of MsgIS136Report values used

Field Name	Type	Description
LastTDMARadioTimeStamp	Datetime2	Timestamp of the last MsgIS136Report message
numTDMAOriginationMsg	int	unused
numTDMADigitalTrafficDe-signation	int	unused
avgiDENRRSSI	real	Average RxLev of the last 8 MobileStatus messages before the end the call
numOfiDENRadioValues	int	Number of MobileStatus values used
LastiDENRadioTimeStamp	Datetime2	Timestamp of the last MobileStatus message
numiDENTchRequest	int	unused
numiDENTchAllocation	int	unused
FailedRegistration	int	1 if there was an unsuccessful registration in UMTS
ReleaseCause	Varchar(100)	RR Cause value of the RR Release
CallMode	Varchar(50)	Mode of the Call (CS, CSFB, VoLTE, SRVCC)
avgLTERSSI	real	Average RSSI of the 8 last LTEMeasurementReport messages before the end of the call
avgLTERSRQ	real	Average RSRQ of the 8 last LTEMeasurementReport messages before the end of the call
avgLTERSRP	real	Average RSRP of the 8 last LTEMeasurementReport messages before the end of the call
avgLTESINR	real	Average SINR0 of the 8 last LTEMeasurementReport messages before the end of the call
avgPUSCHTxPower	real	unused
numOfLTERadioValues	int	Number of LTE radio values from LTE
LastLTERadioTimeStamp	Datetime2	Timestamp of the last MobileStatus message
LastLTEHOTimeStamp	Datetime2	Timestamp of the last LTE Handover (HO)
LastLTEHOType	Varchar(100)	Set to HO_IRAT and HO_INTRALTE
LastLTECause	Varchar(100)	Set to ‘unknown’
numDCCH4GTrial	int	Number of KPI 18000 measurements
numDCCH4GSuccess	int	Number of successful KPI 18000 measurements
numDedicatedBearer	int	Number of KPI 18015 measurements in the call
numDedicatedBearerSuc-cess	int	Number of successful KPI 18015 measurements in the call
numCSFBRRCConnection-Request	int	Number of KPI 10170 measurements in the call
numCSFBEMMExtendedSer-viceRequest	int	Number of KPI 10178 measurements in the call
numCSFBEMMServiceReject	int	Number of Service Reject messages in the call

Field Name	Type	Description
numCSFBRRCRelease	int	Number of successful KPI 10181 measurements and MobilityFromEUTRACommand and RRCConnectionRelease messages in the call
numCSFB4GIRATchange	int	Number of KPI10182 and KPI 38040 measurements in the call
numVoLTEINVITERequest	int	Number of SIP INVITE Request messages in the call
numVoLTEINVITEFailure	int	Number of SIP INVITE Failure messages in the call
numVoLTEINVITETrying	int	Number of SIP INVITE Trying messages in the call
numVoLTEPRACKRequest	int	Number of SIP PRACK Request messages in the call
numVoLTEPRACKFailure	int	Number of SIP PRACK Failure messages in the call
numVoLTEPRACKOK	int	Number of SIP PRACK OK messages in the call
numVoLTEINVITERinging	int	Number of SIP INVITE Ringing messages in the call
numVoLTEINVITEOK	int	Number of SIP INVITE OK messages in the call
numVoLTEByeRequest	int	Number of SIP BYE Requests messages in the call
numVoLTEByeFailure	int	Number of SIP BYE Failure messages in the call
numVoLTEByeOK	int	Number of SIP BYE OK messages in the call
Side	Varchar(50)	Side for which the Call Analysis is running
SessionIdA	bigint	For the B-side it corresponds to the information of the related SessionId of the A-side
JointCode	Varchar(255)	Unused They will hold the analysis result combined from A-side and B-side
JointCodeDescription	Varchar(255)	Unused They will hold the analysis result combined from A-side and B-side
CallTechnologies	Varchar(50)	Unused It will contain the respective combined information for the call. Currently, this combined information is put to the A-Side row in CallMode column

4 Code and CodeDescription Fields

The "sqCallAnalysisCode" stored procedure calculates the call analysis code and the call analysis code description based on the values in the callAnalysis table.

The code field uses the format, *C3.01*, where:

- ‘C’ = Shortcut for ‘Code’
- First number = indication of call status
  (1 = Completed, 2 = Dropped, 3 = Failed, 9 = others)
- Second number = Subcode that describes additional details

4.1 Call Code Description

The codeDescription files decode the Code files into Text.

Table 4-1: Description of Call Codes

Priority	Code	CodeDescription	Description
		Configuration calls are call made to configure, to start, or to stop a scenario. These calls are not taken into account and are not listed in the Report	
	C0.00	Config Call	This was a configuration call. Either to start or to terminate a scenario.
	C0.01	System Release	This is a call that is not a regular call with respect to the defined jobs. For example, config calls, stop calls, or calls that were terminated prematurely due to system constraints.
		Completed Calls. The criterions for a completed call are: successful call setup and successful termination of the call	
	C1.00	Completed, (Disconnect Cause value)	Normal completed call
	C1.10	Completed, Bad Call (Disconnect Cause value)	Normal completed call with a percentage of bad samples above the threshold
	C1.11	Completed, Bad Samples (Disconnect Cause value)	Normal completed call with at least one bad sample (I.LQ < @BadSampleThreshold)
	C1.99	Completed, others	All other completed calls
		Dropped Calls. The criterions for Dropped calls are: not completed calls and successful call setup and unsuccessful call termination	
	C2.02	Dropped, Call Re-established	Dropped but successfully reestablished (disconDirection = 'U' and NumReestablishmentRequests > 0)
	C2.03	Dropped, Call Reestablishment attempt	Dropped and Call reestablishment attempt (NumReestablishmentRequests > 0)

Priority	Code	CodeDescription	Description
	C2.05	Dropped, No Service/Scanning	Dropped call where at least once a No Service or Scanning occurred (NoService > 0 or Initializing > 0)
	C2.06	Dropped, Last HO successfull	Dropped call where the last Handover was successful (lastHoType = 'HO_CMPL')
	C2.07	Dropped, No Handover	Dropped call where no Handover occurred (lastHoType <> 'HO_FAIL' AND lastHoType <> 'HO_CMPL' AND lastHoType <> 'HO_CMD')
	C2.10	Dropped, Disconnect sent (Disconnect Cause value)	Dropped call and disconnect sent by called party (disconDirection = 'U' and callDir = 'B->A')
	C2.11	Dropped, Disconnect received (Disconnect Cause value)	Dropped call and disconnect received (disconDirection = 'D')
	C2.12	Dropped, Disconnect sent by calling party (Disconnect Cause value)	Dropped call and disconnect received (disconDirection = 'U' and callDir = 'A->B')
	C2.20	Dropped, Channel Release received (Releaset Cause value)	Dropped call and Channel Release received (numChnRel > 0)
	C2.21	Dropped, RRCConnectionRelease received	Dropped call call and RRCConnection Release received (numRRCConnRel > 0)
	C2.30	Dropped, Failed Handover (causeValue)	Dropped call with last Handover failed (lastHoType = 'HO_FAIL')
	C2.31	Dropped, Handover uncompleted	Dropped call with last Handover uncompleted (lastHoType = 'HO_CMD')
	C2.32	Dropped, Failed Assignment (causeValue)	Dropped call with last Assignment failed (lastHoType = 'ASS_FAIL')
	C2.33	Dropped, Assignment uncompleted	Dropped calls with uncompleted Assignment (lastHoType = 'ASS_CMD')
	C2.34	Dropped, Failed IRAT-Handover	Dropped call with failed IRAT Handover (lastUMTSHoType = 'HO_IRAT_FAIL')
	C2.40	Dropped, Radio Link Timeout	Dropped call with Radio Link Timeout (RLCExp > 0) This comment can be added to other descriptions without applying the code.
	C2.41	Dropped, High RxQual	Dropped call with average of the last 8 RxQual messages over threshold (avgRxQual >= @poorRxQual) This comment can be added to other descriptions without applying the code.
	C2.42	Dropped, Low RxLev	Dropped call with average of the last 8 RxLev messages below threshold (avgRxLev <= @poorRxLev OR avgBCCHRxLev <= @poorRxLev) This comment can be added to other descriptions without applying the code.

Priority	Code	CodeDescription	Description
	C2.43	Dropped, High BLER'	Dropped call with average of the last 4 BLERmessages above threshold (avgBLER >= @BLERThreshold) This comment can be added to other descriptions without applying the code.
	C2.44	Dropped, Low Total Ec/Io	Dropped call with average of the last 8 aggregated Ec/Io messages below threshold (avgTotEcIo <= @UEEcIoThreshold) This comment can be added to other descriptions without applying the code. (This comment can be added to other descriptions without applying the code)
	C2.45	Dropped, Low UE Rx Power	Dropped call with average of the last 8 UE RxPower messages below threshold (avgUERxPwr <= @UERxLevThreshold) This comment can be added to other descriptions without applying the code.
	C2.46	Dropped, High FER	Dropped call with average of the last 2 FER messages above threshold (avgCDMAFER <= @CDMAFERThreshold) This comment can be added to other descriptions without applying the code.
	C2.47	Dropped, Low Total Ec/Io	Dropped call with average of the last 8 aggregated Ec/Io messages below threshold (avgCDMATotEcIo <= @CDMAECIOThreshold) This comment can be added to other descriptions without applying the code.
	C2.48	Dropped, Low AGC	Dropped call with average of the last 8 AGC messages below threshold (avgCDMAAGC <= @CDMAAGCThreshold) This comment can be added to other descriptions without applying the code.
	C2.49	Dropped, Low IS-136 RxLev	Dropped call with average of the last 8 RxLev messages below threshold (avgTDMARxLev <= @TDMARxLevThreshold) This comment can be added to other descriptions without applying the code.
	C2.50	Dropped, Low iDEN RxLev	Dropped call with average of the last 8 RxLev messages below threshold (avgiDENRRSSI <= @iDENRxLevThreshold) This comment can be added to other descriptions without applying the code.
	C2.98	Dropped, Layer 3 Gaps/ No Layer3 Messages	Dropped call with no or missing Layer 3 Messages (Layer3Messages = 0 or noL3Duration <= @noL3Duration)
	C2.99	Dropped, Others	All other dropped calls
		Failed calls. The criterions for failed calls are: not completed calls and not dropped calls and dial command sent to the device (mobile or fixed device)	

Priority	Code	CodeDescription	Description
	C3.00	Failed, Scanning	Failed call with phone sending at least once the message ‘scanning’ (Initializing > 0) and there was no progress detected
	C3.01	Failed, No Service	Failed call where the no service message occured (NoService > 0)
	C3.10	Failed, Disconnect sent by Called Party (Disconnect Cause value)	Failed call with Disconnect sent by called party (disconDirection = 'U' and callDir <> 'A->B')
	C3.11	Failed, B-Side not correctly connected (Disconnect Cause value)	Failed call where call was connected to wrong party (numconn > 0)
	C3.12	Failed, No Answer (Disconnect Cause value)	Failed call where disconnect was after the alerting (numAlerting > 0)
	C3.13	Failed, Disconnect sent after Progress (Disconnect Cause value)	Failed call where disconnect was after progress (numProgess > 0)
	C3.14	Failed, Disconnect sent by Calling Party (Disconnect Cause value)	Failed call where disconnect was sent by calling party (Dial Timeout)
	C3.15	Failed, Disconnect received by Called Party after Alerting (Disconnect Cause value)	Failed call where disconnect was received by called party after Alerting (callDir <> 'A->B' and numAlerting > 0)(Dial Timeout)
	C3.16	Failed, Disconnect received by Called Party (Disconnect Cause value)	Failed call where disconnect was received by called party (callDir <> 'A->B')(Dial Timeout)
	C3.17	Failed, Disconnect received by Calling Party after Alerting (Disconnect Cause value)	Failed call where disconnect was received by calling party after Alerting (numAlerting > 0)
	C3.18	Failed, Disconnect received by Calling Party (Disconnect Cause value)	Failed call where disconnect was received by calling party (NumAlerting = 0)
	C3.19	Failed, Disconnect not determined (Disconnect Cause value)	Failed call where is not closer determined (numDisconnect > 0)
	C3.20	Failed, Service Reject	Failed call where service was rejected and the connection released (numCMServiceReject > 0)
	C3.21	Failed, Release after Alerting (Release Cause value)	Failed call where a release occurred after alering (numAlerting > 0)
	C3.22	Failed, Release after Progress (Release Cause value)	Failed call where a release occurred after progress (numProgress > 0)
	C3.23	Failed, Release after Service Request (Release Cause value)	Failed call where a release occurred after page response (numCMServiceRequest > 0)
	C3.24	Failed, Release after Page Response (Release Cause value)	Failed call where a release occurred after service request (numPageResponse > 0)
	C3.25	Failed, Release Received by Calling Party (Release Cause value)	Failed call where a the calling party received a release before any progress
	C3.26	Failed, Release Received by Called Party (Release Cause value)	Failed call where a the called party received a release before any progress (callDir <> 'A->B')
	C3.30	Failed, No Progress after Alerting	Failed call where there was no progress after alerting (numAlerting > 0)
	C3.31	Failed, Stop with Progress	Failed call where there was no progress after progress (numProgress > 0)

Priority	Code	CodeDescription	Description
	C3.32	Failed, No Progress after TCH assignment	Failed call where there was no progress after TCH assignment (numTCHSuccess > 0 or numRABSuccess > 0)
	C3.33	Failed, No Progress after Call Confirmed/ Proceeding	Failed call where there was no progress after Confirmed/Proceeding (numCallProceding > 0)
	C3.34	Failed, No Progress after Setup	Failed call where there was no progress after Setup (numSetup > 0)
	C3.35	Failed, No Progress after Service Request/ Page Response	Failed call where there was no progress after Service Request/Page Response (numPageResponse + numCMServiceRequest) > 0
	C3.36	Failed, No Service Request/Page Response	Failed call where there was no Service Request/Page Response (numPageResponse + numCMServiceRequest) = 0
	C3.37	Failed, No Progress after DCCH assignment	Failed call where there was no progress after DCCH assignment ((numDCCH2GSuccess > 0 or numDCCH3GSuccess > 0) or (numImmAss > 0))
	C3.38	Failed, No IMM_ASS for this Random Code	Failed call where there was no immediate assignment was found for the random code in the channel request (CRRandomCode = '' AND numChnReq > 0)
	C3.39	Failed, No DCCH Request	Failed call where there was no DCCH request of the calling party was detected (numDCCH2GTrial = 0 or numDCCH3GTrial = 0) and NumImmAss = 0 and Technology in ('GSM','UMTS','GSM /UMTS') and callDir = 'A->B')
	C3.40	Failed, Radio Link Timeout	Failed call where a Radio Link Timeout occured
	C3.41	Failed, TCH assignment Failed	Failed call where the TCH assignment failed (numTCHTrial> 0 or numRABTrial > 0)
	C3.42	Failed, Service Reject	Failed call where service was rejected and the connection not released (numCMServiceReject > 0)
	C3.43	Failed, DCCH Failed / Rejected	Failed call where the DCCH assignment failed or was rejected (numDCCH2GTrial> 0 or numDCCH3GTrial > 0)
	C3.44	Failed, Registration failed	Failed call where an Registration attempt failed (FailedRegistration > 0)
	C3.50	Failed, No Setup -> Location Updating	Failed call where there is no call set up trial is detected but a Location Updating Process is found (NumLuRequest + numLuAccept + NumLuReject) > 0)
	C3.51	Failed, No Setup -> RA Updating	Failed call where there is no call set up trial is detected but a Routing Area Updating Process is found (NumGMMmessages > 0)
	C3.52	Failed, No Setup -> SMS	Failed call where there is no call set up trial is detected but a SMS process is found (NumSMSmessages > 0)

Priority	Code	CodeDescription	Description
	C3.53	Failed, Called Party was active	Failed B-> A call where the called mobile was in other activities ((numGMMMessages + numMMMessages + numSMSMessages) > 0 or (NumLuRequest + num-LuAccept + NumLuReject) > 0)
	C3.54	Failed, Called Party not reached	Failed B-> A call where the mobile did not react
	C3.60	Failed, No Progress after Service Connect Complete	Failed call where there was no progress after Service Connect Complete (numCDMAServiceConnectCmpl) > 0)
	C3.61	Failed, No Service Connect	Failed call where there was Service Connect message ((numCDMAServiceConnect) =0 and (numCDMAPageResponse + numCDMAOrigina-tionMsg) > 0)
	C3.62	Failed, No Progress after CDMA Origination / Page Response	Failed call where there was no progress after sending the Origination or Page Response message ((numCD-MAPageResponse + numCDMAOriginationMsg) > 0)
	C3.63	Failed, No Origination /Page Response	Failed call where there was no Origination or Page Response message ((numCDMAPageResponse + numCDMAOriginationMsg) = 0)
	C3.70	No Progress after FallBack	Failed call where there was no progess of the call after the fall back to 2G or 3G (numCSFB4GIRATChange > 0)
	C3.71	No FallBack	Failed call where there was no actual fall back to 2G or 3G (numCSFBRRCRelase >0)
	C3.73	EMM Extended Service Request reject	
	C3.75	No Progress after EMM Extended Service Request	Failed call where there was no Progress after the EMM Extended Service Request (numCSFBEMMExtended-ServiceRequest > 0)
		Scanning	Failed call with phone sending at least once the mes-sage ‘scanning’ (Initializing > 0) (This comment is added to other descriptions without applying any code)
		High RxQual	Failed call with average of the last 8 RxQual messages over threshold (avgRxQual >= @poorRxQual) (This comment is added to other descriptions without apply-ing any code)
		Low RxLev	Failed call with average of the last 8 RxLev messages below threshold (avgRxLev <= @poorRxLev OR avgBCCHRxLev <= @poorRxLev) (This comment is added to other descriptions without applying any code)
		High BLER'	Failed call with average of the last 4 BLERmessages above threshold (avgBLER >= @BLERThreshold) (This comment is added to other descriptions without apply-ing any code)
		Low Total Ec/Io	Failed call with average of the last 8 aggregated Ec/Io messages below threshold (avgTotEcIo <= @UEEcIo-Threshold) (This comment is added to other descrip-tions without applying any code)

Priority	Code	CodeDescription	Description
		Low UE Rx Power	Failed call with average of the last 8 UE RxPower messages below threshold (avgUERxPwr <= @UERxLevThreshold) (This comment is added to other descriptions without applying any code)
		High FER	Failed call with average of the last 2 FER messages above threshold (avgCDMAFER <= @CDMAFERThreshold) (This comment is added to other descriptions without applying any code)
		Low Total Ec/Io	Failed call with average of the last 8 aggregated Ec/Io messages below threshold (avgCDMATotEcIo <= @CDMAECIOThreshold) (This comment is added to other descriptions without applying any code)
		Low AGC	Failed call with average of the last 8 AGC messages below threshold (avgCDMAAGC <= @CDMAAGCThreshold) (This comment is added to other descriptions without applying any code)
		Low IS-136 RxLev	Failed call with average of the last 8 RxLev messages below threshold (avgTDMARxLev <= @TDMARxLevThreshold) (This comment can be added to other descriptions without applying the code)
		Low iDEN RxLev	Failed call with average of the last 8 RxLev messages below threshold (avgiDENRRSSI <= @iDENRxLevThreshold) (This comment can be added to other descriptions without applying the code)
		Calling Party: (CallCause)	Failed call where the calling party determined a particular call cause (callDir = 'B->A') (This comment is added to other descriptions without applying any code)
	C3.98	Failed, No Layer 3 Messages/ Layer3 Gaps	Failed call with no or missing Layer 3 Messages (Layer3Messages = 0 or noL3Duration <= @noL3Duration)
	C3.99	Failed, others	All other failed calls

5 Generating a Call Analysis Report

After you run the Call Analysis script, you can generate the `CallAnalysis.XLS` report file in Excel format. In the report, the calls are grouped by the "Operator", "Call Status", and "Code Description" fields. If you want, you can use the "Technology" field to further distinguish between the different technologies.

► To generate the report file, click "Reports" on the "Tools" menu, point to "Voice", and then click "Call Analysis".

				Operator	
				Sunrise	
codeGroup	code	codeDescription	Data	L->M	M-
Completed	C1.00	Completed, Normal call clearing	Count	39	
			%	97.50%	
		Completed,	Count		
			%	0.00%	
	C1.10	Completed, Bad Call, Normal call clearing	Count		
			%	0.00%	
Completed Count				**39**	
Completed %				**97.50%**	
Dropped	C2.10	Dropped, Disconnect sent (Normal call clearing-Normal-user)	Count	1	
			%	2.50%	
	C2.21	Dropped, RRCConnectionRelease received	Count		
			%	0.00%	
	C2.40	Dropped, Radio Link Timeout, Call Re-Establishment attempt	Count		
			%	0.00%	
Dropped Count				**1**	
Dropped %				**2.50%**	
Failed	C3.11	Failed, B-Side not correctly connected (Silence)	Count		
			%	0.00%	
	C3.15	Failed, Disconnect received by Called Party after Alerting (Normal call clearing) ; Calling Party: Silence	Count		
			%	0.00%	
	C3.33	Failed, No Progress after Call Confirmed/Proceeding	Count		
			%	0.00%	
	C3.37	Failed, No Progress after DCCH assignment	Count		
			%	0.00%	
	C3.39	Failed, No DCCH Request , High BLER, Low UE Rx Power	Count		
			%	0.00%	
	C3.41	Failed, TCH assignment Failed	Count		
			%	0.00%	
	C3.50	Failed, No Setup -> Location Updating	Count		
			%	0.00%	
	C3.70	Failed, No Progress after FallBack	Count		
			%	0.00%	

Figure 5-1: Call Analysis Report Example